

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer Examination – 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: IV

Subject Code & Name: BTCVC405 Hydraulics - II

Max Marks: 60

Date: 24/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- | | | | |
|----|---|------------|---|
| A) | Explain open channel flow and state difference between open channel flow and pipe flow. | Understand | 6 |
| B) | Derive an expression for discharge through triangular notch with neat diagram. | CO1 | 6 |
| C) | A reservoir $4.65 \times 10^4 \text{ m}^2$ in area is to be controlled by a rectangular weir with its crest level at El. 30 m. it is intended to provide such a length of weir that will lower the water level from El. 31.2 m to El. 30.6 m in half an hour time. Determine the length of the weir. The discharge over the weir is given by the formula $Q = 1.9 LH^{3/2}$, where Q is discharge in cumec, L is crest length in meters and H is head over the weir in meters. | CO1 | 6 |

Q.2 Solve Any Two of the following.

12

- | | | | |
|----|--|----------|---|
| A) | Define the term most economical section of channel. Derive the conditions for the most economical triangular channel section? | CO1 | 6 |
| B) | Define specific energy and give detailed explanation about specific energy curve with neat labelled sketch. | Remember | 6 |
| C) | Design the most economical trapezoidal canal section having one side vertical and other side slope 1.5 H:1 V to carry the discharge of $10 \text{ m}^3/\text{s}$. Take Manning's coefficient $n = 0.015$ and bed slope = $1/3500$. | Create | 6 |

Q. 3 Solve Any Two of the following.

12

- | | | | |
|----|---|-----|---|
| A) | Derive the dynamic equation for gradually varied flow with assumptions in | CO2 | 6 |
|----|---|-----|---|

the form:
$$\frac{dy}{dx} = \frac{S_0 - S_f}{1 - F_r^2}$$

- B) With usual notation in hydraulic jump, prove the following relation for hydraulic jump in rectangular channel CO2 6

$$\frac{y_2}{y_1} = \frac{1}{2} \left[-1 + \sqrt{1 + 8F_{r1}^2} \right]$$

- C) A jet of water 5 cm in diameter having a velocity of 25 m/s strikes normally a smooth flat plate. Determine the thrust when CO3 6
- i. The plate is stationary.
 - ii. The plate is moving with a velocity of 5 m/s away from jet along the line of jet.
- What is the work done per second by the jet in each case?

Q.4 Solve Any Two of the following. 12

- A) Enlist types of turbines. Describe components of Francis Turbine with neat labelled sketch. Remember 6
- B) Discuss the criteria considered for the selection of turbines. Understand 6
- C) Give description about draft tube and its function. Remember 6

Q. 5 Solve Any Two of the following. 12

- A) What are different component parts of a centrifugal pump? Explain their function with neat sketch. Remember 6
- B) Why priming is necessary for centrifugal pump? Enlist & describe priming devices used for priming. Understand 6
- C) Explain i) Submersible pump Remember 6
ii) Multistage pump

*** End ***